Purdue University

College Of Science

STAT 355

Project Report

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Honor Pledge

“As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - We are Purdue.”

Introduction:

The raw dataset that I chose is from kaggle. It is Web Scrapped through Seleniun and contains unlabelled text data of around 9000 Netflix Shows and Movies along with full details like Cast, Release Year, Rating, Description, etc. It has 12 columns/variables which are show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description. I chose this dataset specifically because I wanted to work with some data of an organization or application that I am familiar with or have used in the past. Netflix, being my go-to streaming service for many years, was the perfect choice.

Part 1:

Variables:

As I mentioned earlier, the dataset deals with 12 columns and upon further investigation through R, it has 8807 rows. Lets dive a little into specifics and characteristics of each column.

1. show_id: A character variable that likely serves as a unique identifier for each show or movie.

2. type: Indicates whether the entry is a "Movie" or a "TV Show".

3. title: Contains the title of the show or movie.

4. director: Lists the director(s) of the show or movie.

5. cast: Contains the names of actors and actresses who appear in the show or movie.

6. country: Specifies the country or countries where the show or movie was produced or filmed.

7. date_added: Indicates the date when the show or movie was added to the Netflix platform.

8. release_year: Denotes the year when the show or movie was originally released.

9. rating: Indicates the content rating assigned to the show or movie.

10. duration: Specifies the duration of the show or movie, either in terms of runtime (for movies) or number of seasons/episodes (for TV shows).

11. listed_in: Contains categorical labels or genres assigned to the show or movie.

12. description: Contains a brief description or summary of the show or movie.

Structure of the dataset:

Using str(netflix_data), I was able to find out the initial structure of the dataset and through some initial exploration, I found the following information:

- The type variable shows a distribution between "Movie" and "TV Show".
- The director variable has some missing values.
- The country variable also has some missing values.
- The date_added variable needs to be converted to Date format for further analysis.
- The release_year variable spans a range of years, indicating the diversity in release dates of the content.
- The rating variable contains different content ratings.
- The duration variable needs to be processed to extract numeric values for further analysis.
- The listed_in variable contains multiple genres separated by commas.
- The description variable contains text data.

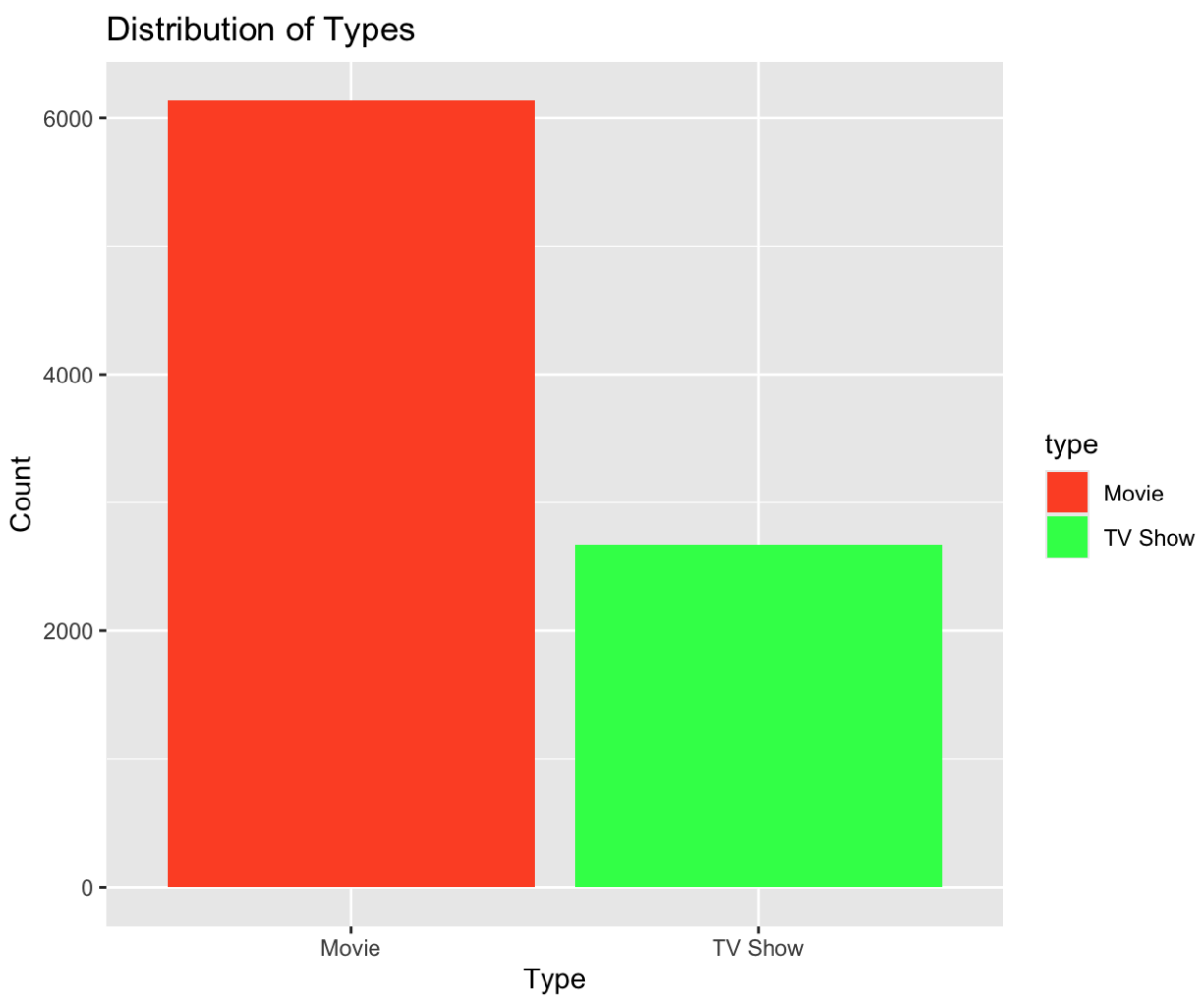
```
> str(netflix_data)
'data.frame':   8807 obs. of  12 variables:
 $ show_id     : chr  "s1" "s2" "s3" "s4" ...
 $ type        : chr  "Movie" "TV Show" "TV Show" "TV Show" ...
 $ title       : chr  "Dick Johnson Is Dead" "Blood & Water" "Ganglands" "Jailbirds New Orleans" ...
 $ director    : chr  "Kirsten Johnson" "" "Julien Leclercq" "" ...
 $ cast        : chr  "" "Ama Qamata, Khosi Ngema, Gail Mabalane, Thabang Molaba, Dillon Windvogel, Natasha Thahane, Arno Greeff, Xolile "| __truncated__ "Sami Bouajila, Tracy Gotoas, Samuel Jouy,
Nabiha Akkari, Sofia Lesaffre, Salim Kechiouche, Noureddine Farihi, G"| __truncated__ "" ...
 $ country     : chr  "United States" "South Africa" "" "" ...
 $ date_added  : chr  "September 25, 2021" "September 24, 2021" "September 24, 2021" "September 24, 2021" ...
 $ release_year: int  2020 2021 2021 2021 2021 2021 2021 1993 2021 2021 ...
 $ rating      : chr  "PG-13" "TV-MA" "TV-MA" "TV-MA" ...
 $ duration    : chr  "90 min" "2 Seasons" "1 Season" "1 Season" ...
 $ listed_in   : chr  "Documentaries" "International TV Shows, TV Dramas, TV Mysteries" "Crime TV Shows, International TV Shows, TV Action & Adventure" "Docuseries, Reality TV" ...
 $ description : chr  "As her father nears the end of his life, filmmaker Kirsten Johnson stages his death in inventive and comical wa"| __truncated__ "After crossing paths at a party, a Cape Town
teen sets out to prove whether a private-school swimming star is h"| __truncated__ "To protect his family from a powerful drug lord, skilled thief Mehdi and his expert team of robbers are pulled
"| __truncated__ "Feuds, flirtations and toilet talk go down among the incarcerated women at the Orleans Justice Center in New Or"| __truncated__ ...
```

Further Analysis:

As show_id is just a alphanumeric code to distinguish between the netflix movies/shows, it does not need any further exploration and due to this we will start off with **type** variable. With some help from R, we can see that there are 6131 movies and 2676 tv shows in this dataset.

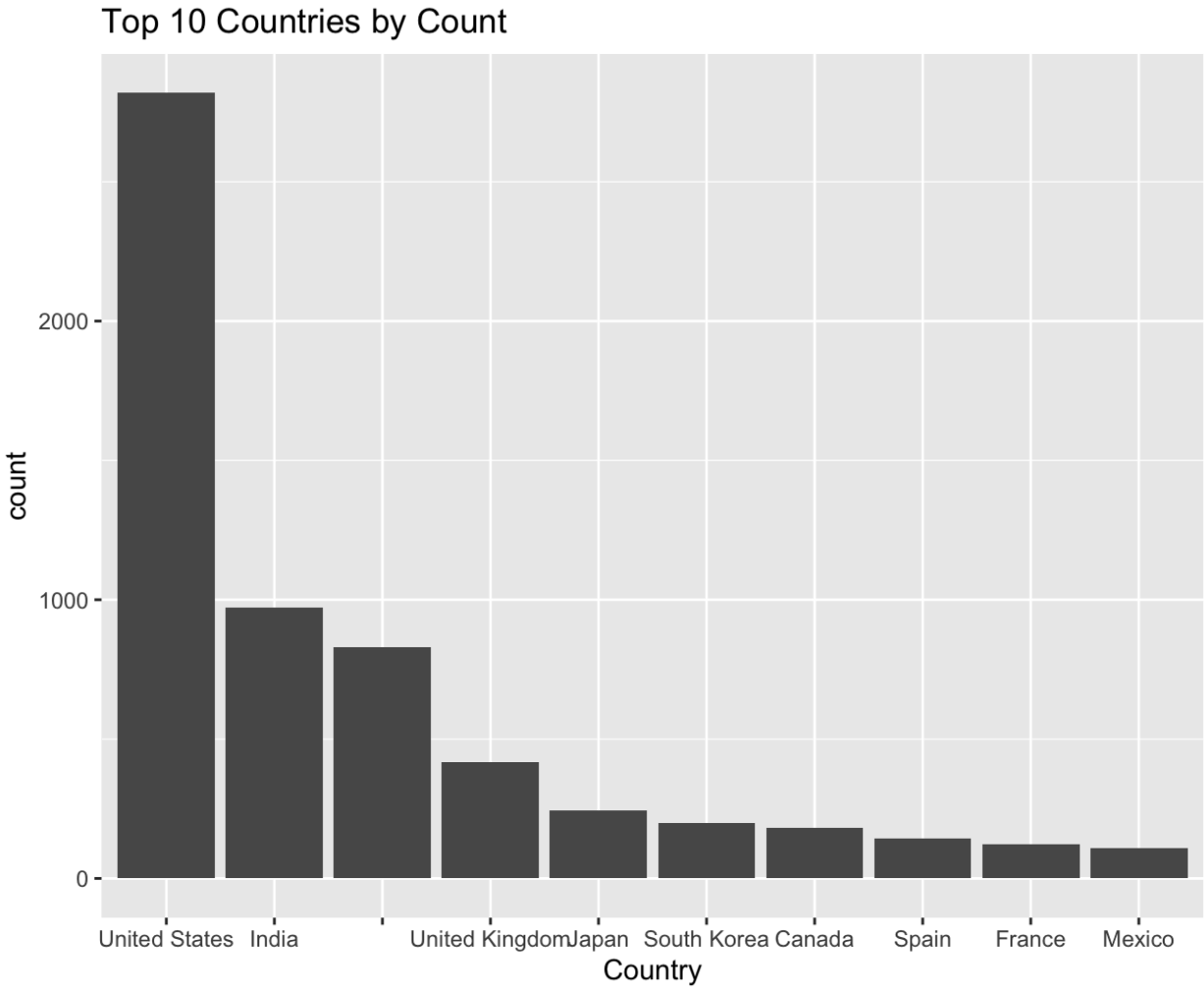
```
table(netflix_data$type)

Movie TV Show
 6131    2676
```

To get a more visualized view, I also created a bar plot for type with some custom colors for better visualization.



As seen in the graph, there are more than twice the amount of movies on Netflix than there are tv shows.

Now, lets move on to the **country** variable. With the help of R, I was able to make a plot which shows the countries having the most movies made or produced. The result is the following:



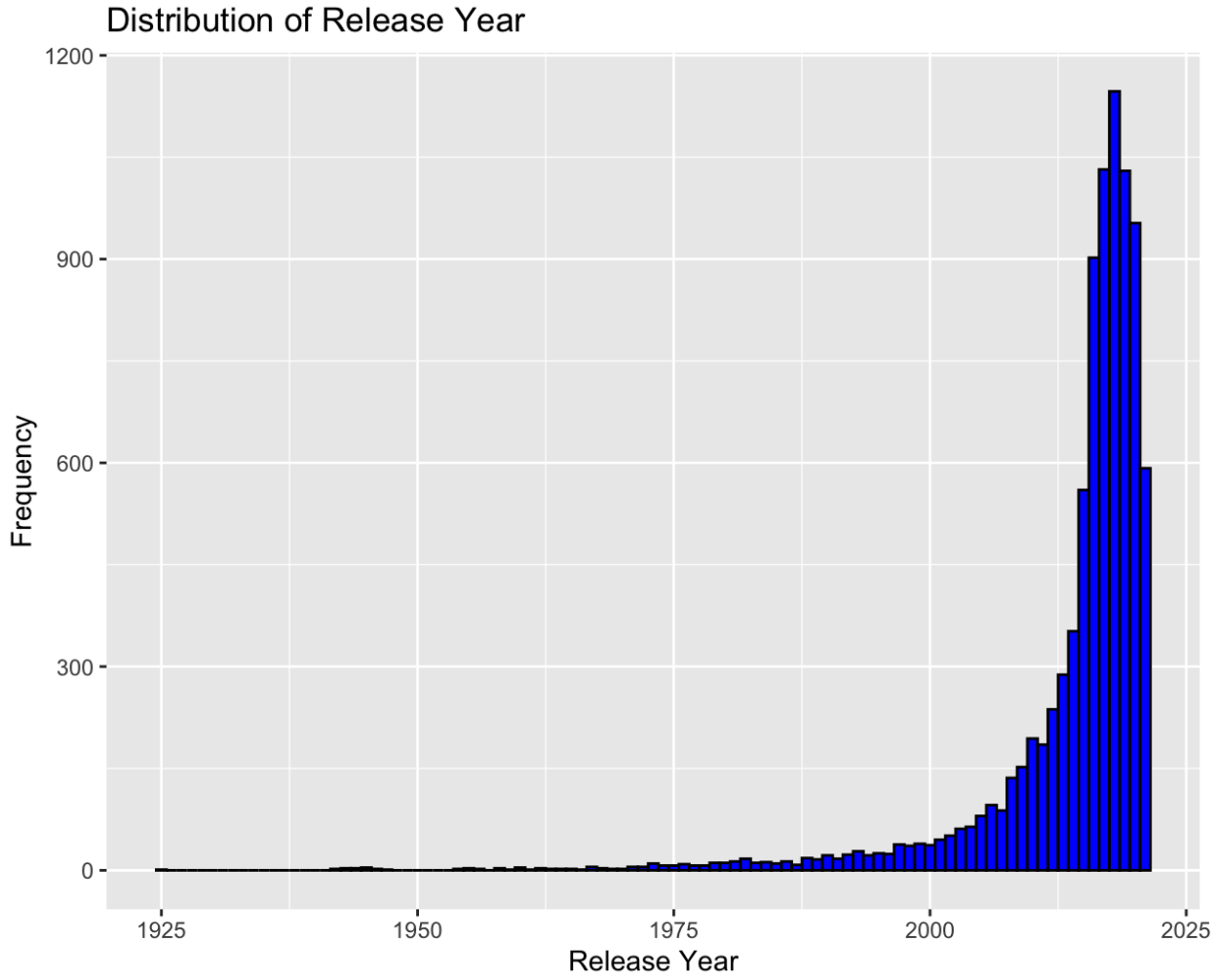
With the help of the plot, we can see that United States has the most content made/ produced on Netflix with around 2800 content followed up by India with almost 1000. You may have noticed that the third value doesnt have an x-axis label and it means that around 800 of the content in the dataset has missing values. So, from this plot we can say that United States has te most content produced on Netflix.

Moving on to **release_year** variable, we can use the summary() function to get a quick summary of the variable.

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1925    2013    2017    2014    2019    2021
```

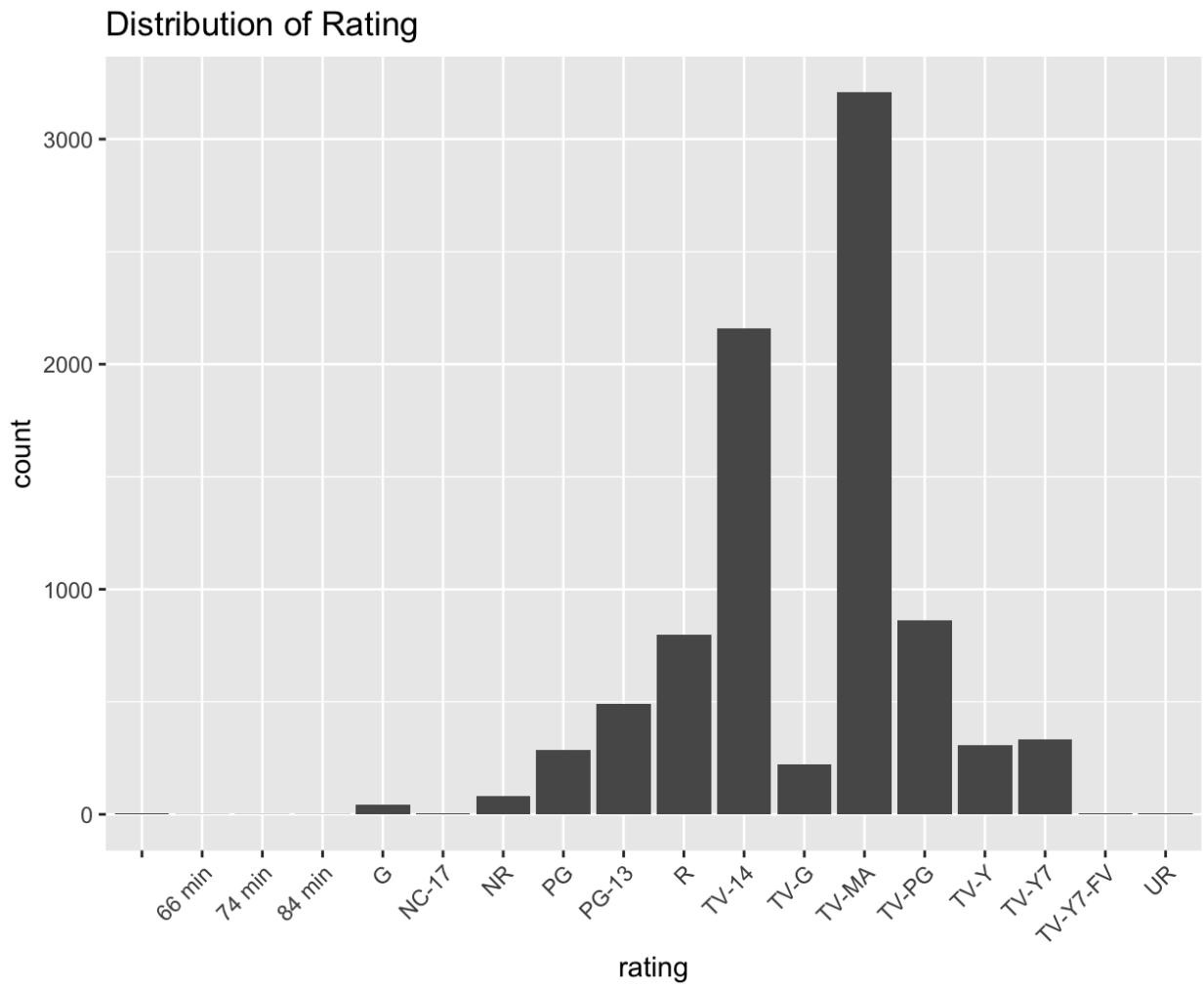
From this information, we can conclude that the oldest content on Netflix is from 1925 with the mean and median being 2014 and 2017 respectively. This also helps us conclude that, on average, the content on Netflix is likely to be produced near the year

2014. To further our investigation, I also created a histogram so that we can visualize the characteristics of release_year.



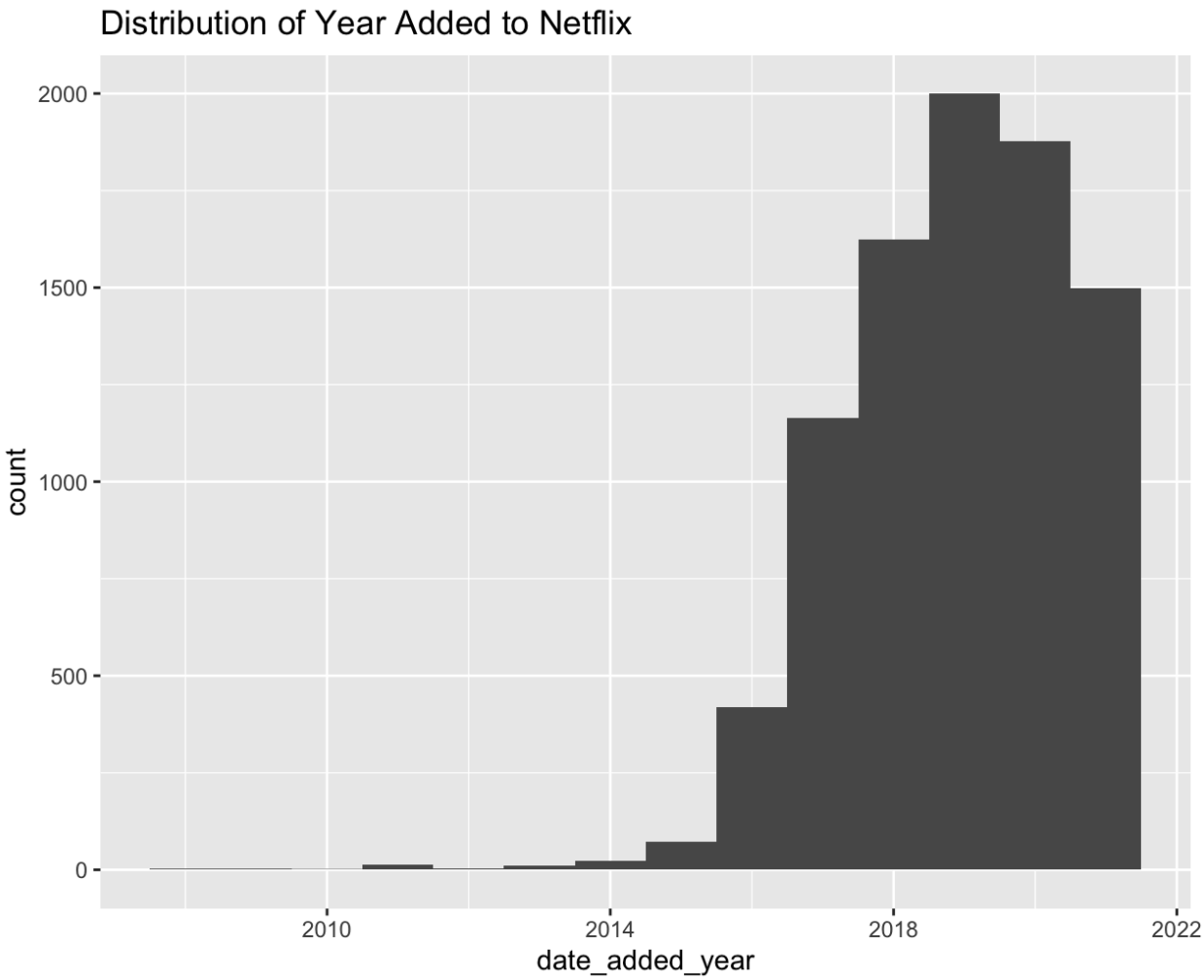
From the histogram, we can see that from around 1925 to just before 2000, there are very few bars, indicating a low frequency or count of releases. This suggests that Netflix has relatively less content that was released during this period. Also, there's a significant increase in the height of the bars starting just before the year 2000, indicating an increase in the number of releases per year. This could be due to the growth of the film and television industry during this period, or it could reflect Netflix's strategy of adding more recent content to its platform. The bars reach their maximum height close to the year 2020, showing that most content was released around this time. This suggests that Netflix has a large amount of content that was released in the last few years. The graph is also left skewed which confirms the fact we discussed earlier that the mean is smaller than the median.

To understand more about the **ratings** variable, I made a plot to compare the different ratings and the amount of movies who had the same rating.



From the plot we can see that tallest bar corresponds to the “TV-MA” rating indicating it has the highest count among all ratings suggesting that a significant portion of Netflix content is intended for mature audiences. The second tallest bar corresponds to “TV-14” which indicates that there is also a substantial amount of content that is suitable for children aged 14 and above. Other ratings like R, PG-13, and PG have significantly lower counts compared to “TV-MA” and “TV-14”, indicating that there is less content with these ratings on Netflix.

Moving on to **date_added** variable, we use the help of a plot to further our investigation.



We can see that, according to this dataset, there were very few additions to Netflix that were not their own production material (also called Netflix Originals) up until 2013-2014. There's a significant increase in content addition observed from 2015 onwards, peaking around 2019. This suggests that Netflix has been adding more and more content each year, with the most content added around 2019. After the peak in 2019, there's a slight decrease in the count of content added. This could be due to various factors such as changes in Netflix's content strategy, availability of new content, or market conditions. We can also see that the last year the content was added was 2019 suggesting that the data is of Netflix until 2019 and more specifically 25th September, 2019.

Part 2:

1. Research Question: Is the average duration of movies significantly different from 100 minutes?

The reason why I find this question helpful and interesting is because it can help understand the typical length of a movie on Netflix. If the average duration is significantly different from 100 minutes, it could suggest viewer preferences for shorter or longer movies.

Method: We can use a one-sample t-test to compare the average movie duration to the hypothesized value of 100 minutes. This test is appropriate because movie duration is a continuous variable and we're comparing the sample mean to a known value.

Assumptions: The test assumes that the durations of movies are normally distributed and that the sample of movies is randomly selected and independent.

We will be using the significance level of 0.05.

After filtering the type column for just movies and converting the duration column of the movies into integer after removing the "mins" in it, I performed the one-sample t-test. Hypotheses:

- Null Hypothesis (H0): The average duration of movies on Netflix is 100 minutes. This is the hypothesis that there is no effect or difference, and it is assumed to be true until we have enough evidence to decide otherwise.
- Alternative Hypothesis (H1): The average duration of movies on Netflix is not 100 minutes. This is the hypothesis that there is an effect or difference.

Upon performing the t-test, these are the results:

```
        One Sample t-test

data:  movies$duration_numeric
t = -1.1699, df = 6127, p-value = 0.2421
alternative hypothesis: true mean is not equal to 100
95 percent confidence interval:
  98.86872 100.28565
sample estimates:
mean of x
 99.57719
```

- t-value: The t-value is -1.1699. The t-value is negative, indicating that the sample mean is less than the hypothesized mean (100 minutes).

- Degrees of Freedom (df): The degrees of freedom for this test is 6127, which is the number of movies minus 1.

- p-value: The p-value is 0.2421. This is greater than the common significance level of 0.05. Therefore, we fail to reject the null hypothesis. This means that we do not have enough evidence to conclude that the average duration of movies on Netflix is significantly different from 100 minutes.

- Confidence Interval: The 95% confidence interval for the true average duration of movies on Netflix is between 98.86872 minutes and 100.28565 minutes. This means that we are 95% confident that the true average duration of movies on Netflix is within this interval.

- Sample Estimate: The sample mean (or estimate of the true mean) is 99.57719 minutes. This is the average duration of movies in the sample.

The p-value is greater than 0.05, so we do not reject the null hypothesismeaning that we don't have enough evidence to say that the average duration of movies on Netflix is significantly different from 100 minutes.

The confidence interval I calculated is 98.86872, 100.28565. This is a 95% confidence interval, which means that if we were to repeat the experiment many times, we would expect the true population parameter to fall within this interval 95% of the time. Since this interval includes 100 minutes, it's consistent with our hypothesis test result that we do not reject the null hypothesis.

In conclusion, both the hypothesis test and the confidence interval suggest that we do not have enough evidence to say that the average duration of movies on Netflix is significantly different from 100 minutes.

Further Analysis:

Netflix's Content Strategy: The average duration of movies on Netflix is approximately 99.58 minutes, which is not significantly different from 100 minutes and this suggests that Netflix's strategy might be to provide movies that are around 1 hour and 40 minutes long, which could be a sweet spot for keeping viewers engaged without overwhelming them with excessively long content.

Viewer Preferences: The fact that the average movie duration is close to 100 minutes might also reflect viewer preferences and it's possible that viewers prefer movies that are around this length, leading to higher viewership and engagement for movies of this duration.

Content Production and Acquiring: The average movie duration could also be influenced by factors related to content production and acquisition. For example, it might be more cost-effective to produce or acquire movies of this length, or it could be that the majority of available movies in the market are around this duration.

Variety in Content: While the average duration is around 100 minutes, the confidence interval (98.87 minutes to 100.29 minutes) suggests that there is variation in movie durations on Netflix. This indicates that Netflix offers a small variety of content to cater to different viewer preferences for movie length.

2. Research Question: Is there a significant difference in the number of seasons for TV shows across different content ratings (like TV-MA, TV-14, etc.)?

The reason why I find this question interesting and helpful is because it can help us understand if certain content ratings are associated with longer or shorter TV shows. For example, TV shows rated for mature audiences (TV-MA) might have more seasons than those suitable for a general audience (G).

Method: We can use ANOVA to compare the means of TV show seasons across different content ratings. This test is appropriate because the number of seasons is a continuous variable and we're comparing the means across more than two groups (content ratings).

Assumptions: The test assumes that the number of seasons within each content rating group are normally distributed, that the variances of the number of seasons in each group are equal (homogeneity of variances), and that the observations are independent.

Hypotheses:

- Null Hypothesis (H0): The average number of seasons is the same across all content ratings. This means that the content rating does not have an effect on the number of seasons a TV show has.
- Alternative Hypothesis (H1): The average number of seasons is not the same across all content ratings. This means that the content rating does have an effect on the number of seasons a TV show has.

Again, we will be suing the default value of significance level in R, which is 0.05.

After performing the ANOVA test, I received the following results:

```
              Df Sum Sq Mean Sq F value Pr(>F)
rating         9     50   5.601   2.245  0.017 *
Residuals   2666   6651   2.495
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- Degrees of Freedom (Df): The degrees of freedom for the rating variable is 9, and for the residuals is 2666. The degrees of freedom for a variable is the number of values that are free to vary, which is the number of levels in the variable minus 1.

- Sum Sq: This is the sum of squares, which is a measure of the total variability in the data. For the rating variable, the sum of squares is 50, and for the residuals, it’s 6651.

- Mean Sq: This is the mean square, which is the sum of squares divided by the degrees of freedom. For the rating variable, the mean square is 5.601, and for the residuals, it’s 2.495.

- F value: The F value is a statistic that is used to decide whether to reject the null hypothesis. It’s calculated as the ratio of the mean square for the variable to the mean square of the residuals. The F value for the rating variable is 2.245.

- Pr(>F): This is the p-value associated with the F statistic. It’s the probability of obtaining the observed data (or data more extreme) if the null hypothesis is true. The p-value for the rating variable is 0.017.

Since the p-value is less than the common significance level of 0.05, we reject the null hypothesis. This means that we have enough evidence to conclude that there is a significant difference in the average number of seasons for TV shows across different content ratings. The asterisk next to the p-value indicates the level of significance. In this case, a single asterisk represents a significance level of 0.05.

In conclusion, based on this analysis, there is a significant difference in the average number of seasons for TV shows across different content ratings on Netflix. This could suggest that certain content ratings are associated with longer or shorter TV shows.

Content Variety: The difference in the number of seasons across content ratings suggests that Netflix offers a wide variety of content to cater to different viewer preferences and that some viewers might prefer shorter series with fewer seasons, while others might prefer longer series with more seasons.

Target Audience: The content rating of a TV show often reflects its target audience. For example, shows with a “TV-MA” rating are intended for mature audiences, while shows with a “TV-G” rating are suitable for all ages. The significant difference in the number of seasons across content ratings could suggest that Netflix’s content strategy is tailored to the viewing habits and preferences of these different audience/viewers.

Content Production and Acquiring: The number of seasons a TV show has can also reflect decisions about content production and acquisition meaning that shows that are popular and successful often have more seasons and, thus, the difference in the number of seasons across content ratings could also suggest that Netflix invests more in certain types of content.

3. Research Question: Is there a relationship between the release year of a movie and its duration?

The reason why this question is interesting is because it can help understand if movies have been getting longer or shorter over the years. For example, if there is a positive correlation between release year and movie duration, it could suggest that movies have been getting longer over time.

Method: We can use correlation to measure the strength and direction of the relationship between release year and movie duration. If there is a significant correlation, we can use simple linear regression to model this relationship and predict movie duration based on release year.

Assumptions: Both correlation and simple linear regression assume that there is a linear relationship between the two variables. Regression also assumes that the residuals are normally distributed and have constant variance, and that the observations are independent.

After calculating the correlation (after omitting the pairs with missing values) and performing the regression, these are the results:

```
> print(correlation)
[1] -0.2062847
```

The correlation between release_year and duration_numeric is -0.2062847. This is a negative correlation, which means that as the release year increases (i.e., for more recent movies), the duration of the movies tends to decrease, and vice versa. However, the correlation is not very strong since it's relatively close to 0.

```
> summary(regression_result)

Call:
lm(formula = duration_numeric ~ release_year, data = movies)

Residuals:
     Min       1Q   Median       3Q      Max 
-124.454  -13.062   -0.636   14.715  215.364 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)  1313.21827   73.55242   17.85   <2e-16 ***
release_year   -0.60287    0.03654  -16.50   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 27.68 on 6126 degrees of freedom
  (3 observations deleted due to missingness)
Multiple R-squared:  0.04255,	Adjusted R-squared:  0.0424 
F-statistic: 272.3 on 1 and 6126 DF,  p-value: < 2.2e-16
```

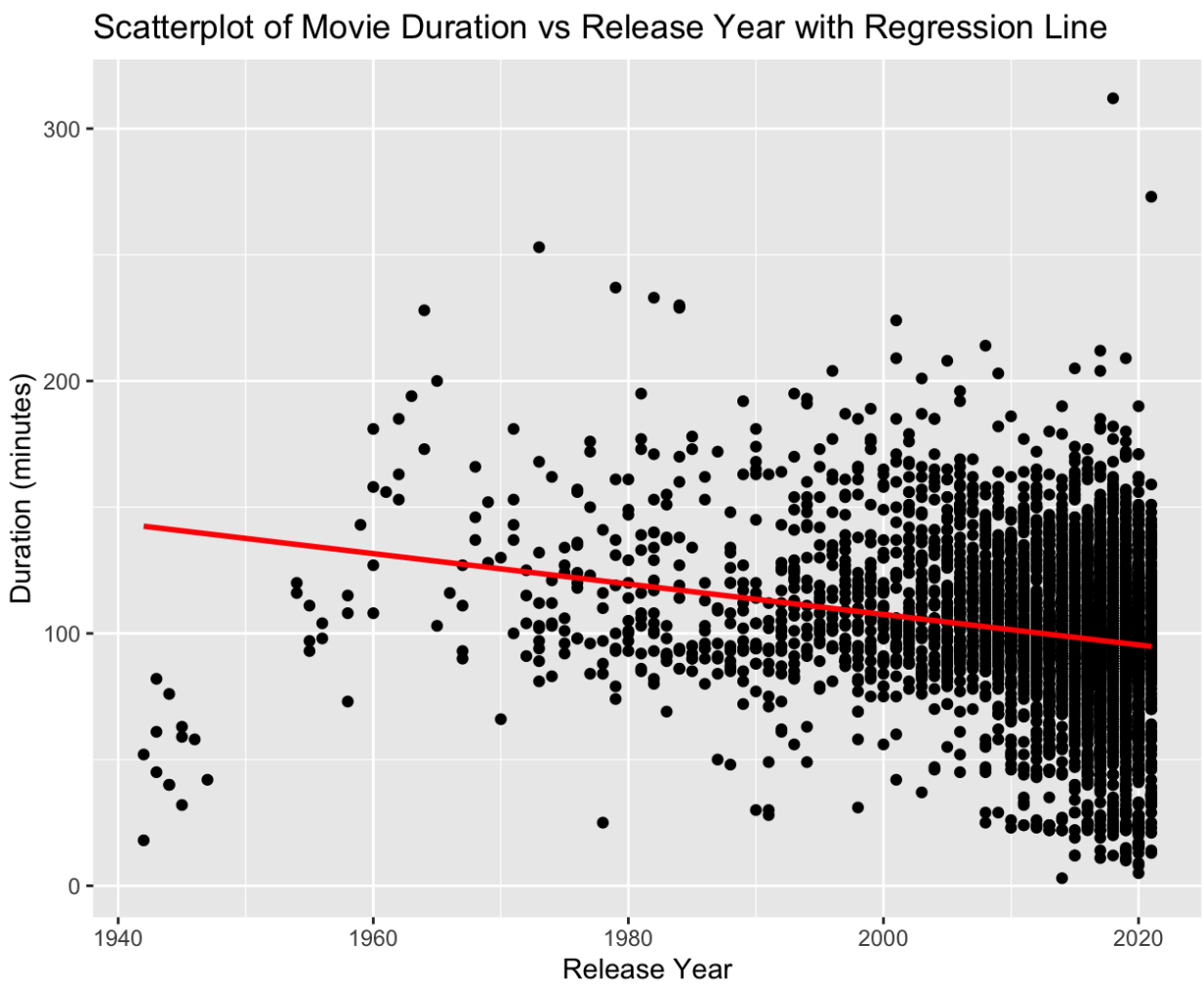
Coefficients: The intercept is 1313.21827 and the slope (coefficient for release_year) is -0.60287. This means that for each additional year, the duration of the movie decreases by approximately 0.60 minutes, on average. The p-values for both the intercept and release_year are less than 0.05 (in fact, they're much smaller), which means that both coefficients are statistically significant.

The R-squared value is 0.04255, which means that approximately 4.255% of the variability in movie duration can be explained by the release year. This is a relatively low value, indicating that release year alone does not explain much of the variability in movie duration.

The F-statistic tests the overall significance of the regression model. The p-value associated with the F-statistic is less than 0.05, indicating that the model is statistically significant.

In conclusion, there is a statistically significant relationship between the release year of a movie and its duration, but the relationship is weak. The negative correlation suggests that more recent movies tend to be shorter, but the release year explains only a small portion of the variability in movie duration. Other factors not included in this model likely also impact movie duration.

To better understand the relationship, I created a scatterplot which can be used to visualize the relationship between the release year and movie duration. Each point represents a movie, with the x-coordinate representing the release year and the y-coordinate representing the duration. I also added a regression line to the scatterplot to visualize the linear regression model.



From the scatterplot, it's clear that there is a negative correlation between the release year and the duration of movies, which is consistent with the correlation value of -0.2062847 that I calculated. This means that as the release year increases (i.e. more recent movies), the duration of the movies tends to decrease.

The red line in the plot is the line of best fit from the linear regression analysis and the negative slope of the line confirms the negative correlation between the release year and movie duration.

However, it's important to note that the points are quite scattered around the regression line, indicating a lot of variability in movie durations that isn't explained by the release year alone. This is reflected in the relatively low R-squared value from the regression analysis, which suggests that only about 4.255% of the variability in movie duration can be explained by the release year.